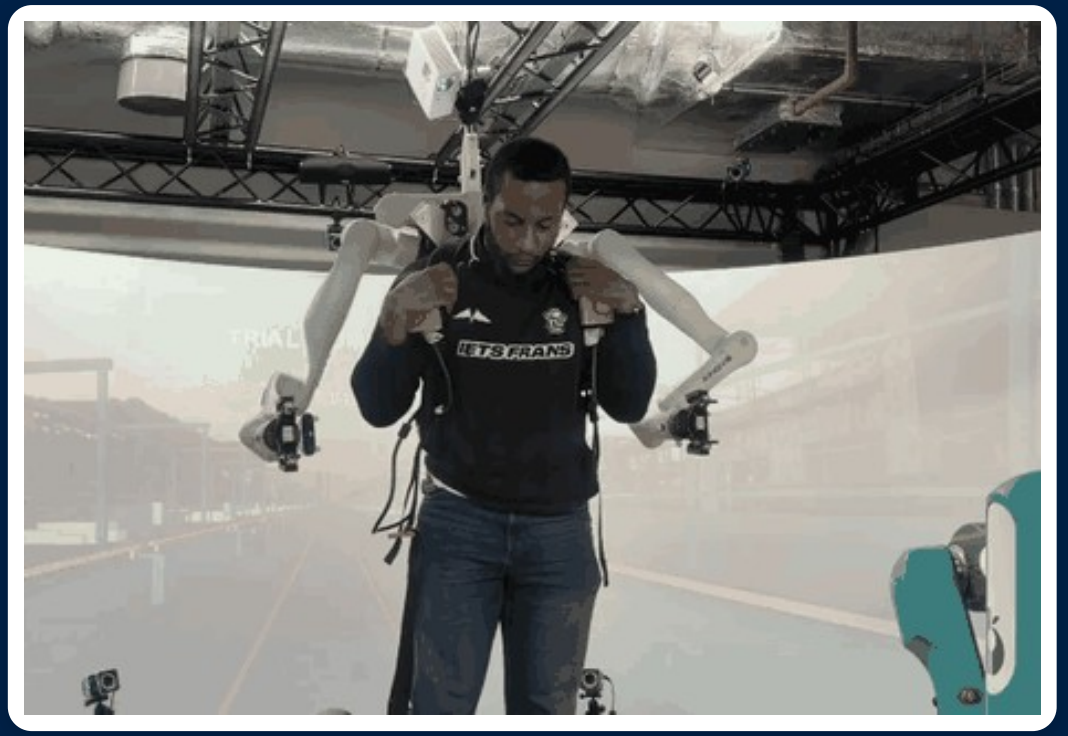
Imperial College London

World-Stable Supernumerary Effectors for Human Augmentation Under Locomotion-Scale Base Motion

Keeping a wearable robot's hands still while the wearer walks

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MSc Human and Biological Robotics
Imperial College London · MUVE Lab · 2026



Watch the hands, not the elbows.

Every step moves your extra hands



63–68 mm

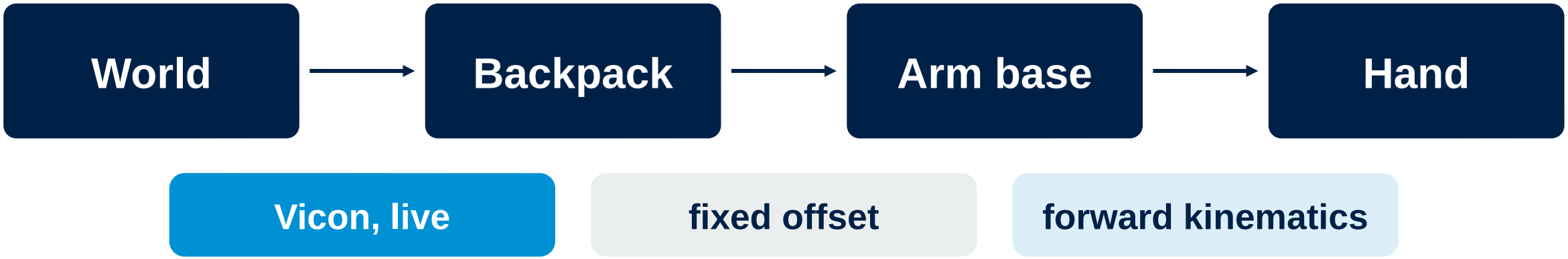
hand motion, arm locked (RMS)

~1.8 Hz

step rate

Goal: hands fixed in the room, not the body

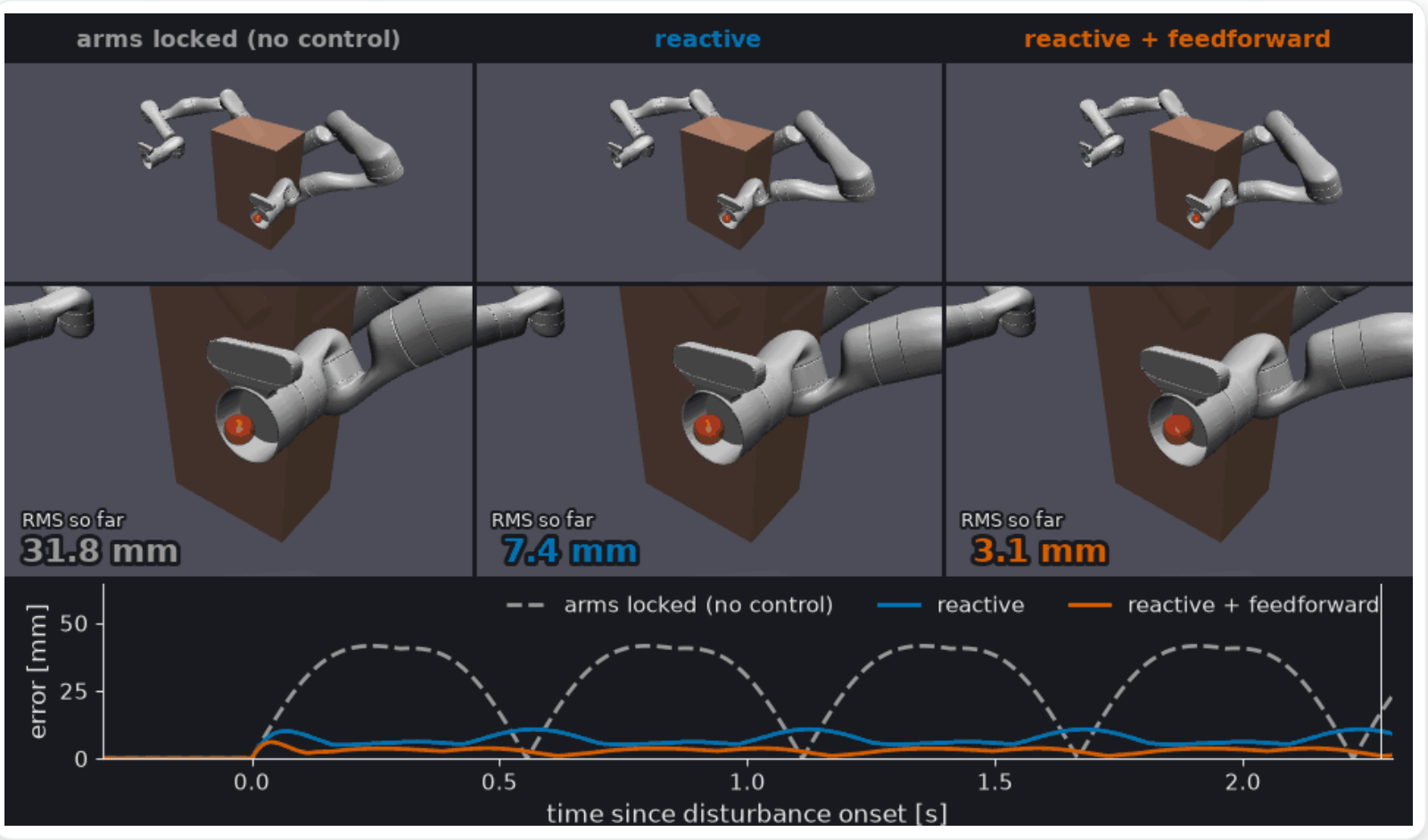
Measure the backpack, cancel it through the arm



$$\dot{q} = J^{+}_{\lambda}\,(K_p \cdot e + K_d \cdot \dot{e})$$

room-frame error → joint speeds ▲ velocity term

In simulation: my controller in a physics engine



Hand error, RMS over 8 s

31.0 mm

arms locked

6.7 mm

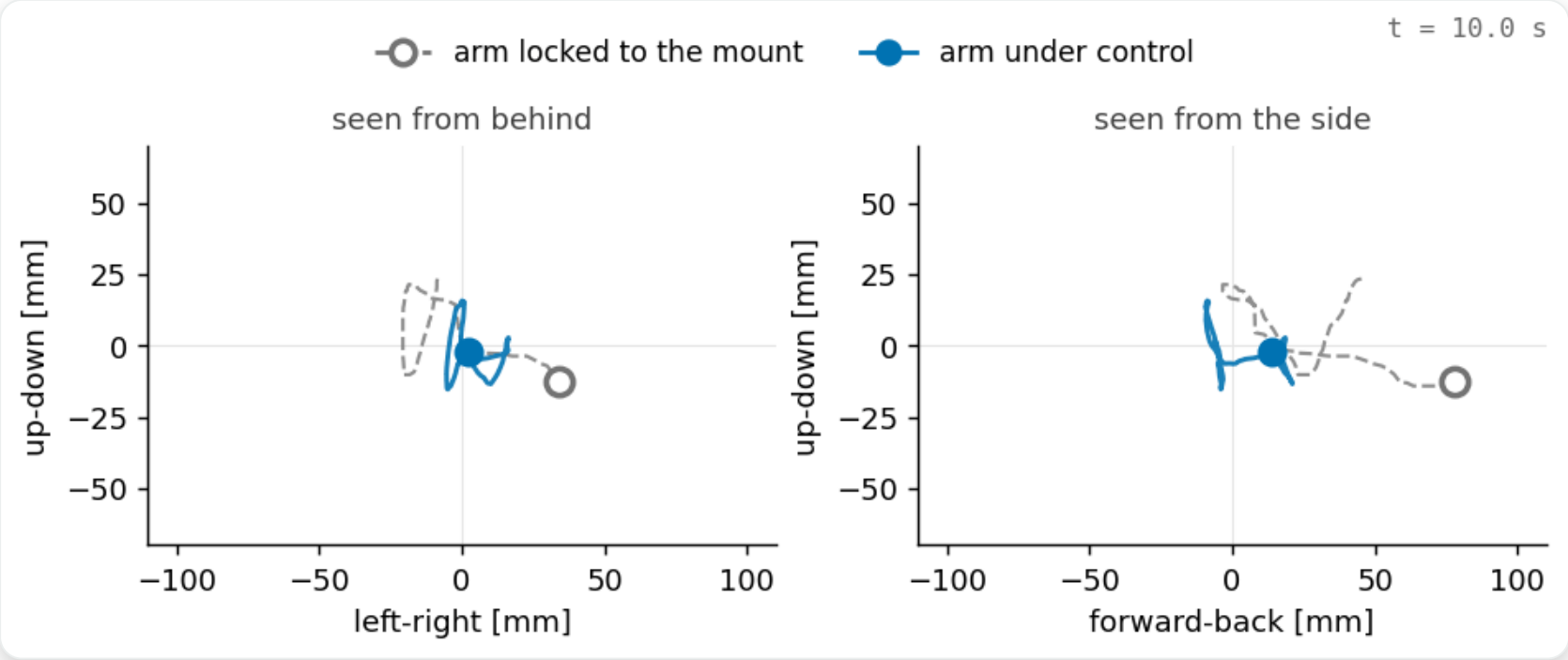
reactive

2.6 mm

reactive + feedforward

On real wearers: about 2/3 of walking motion removed

Trials: a 400 Hz controller on the lab machine. Analysis: mine.



71%

0.5 m/s

63%

1.0 m/s

56%

1.5 m/s

n = 6 (4 at 1.5 m/s)

More velocity gain barely helped. Why?

One-wearer pilot: K_d 0.8 → 2.6. Motion reaching the hand near 1 Hz fell:

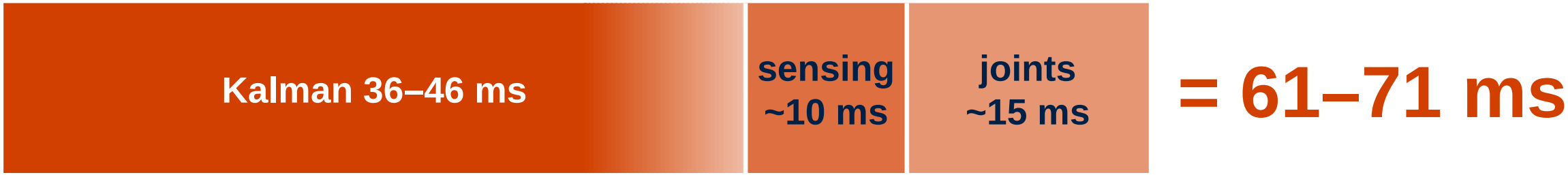
38%

predicted

9%

measured

Best explanation: the velocity arrives 61–71 ms late



Velocity error: $2 \cdot \sin(\omega d/2) \approx 70\%$ at 1.8 Hz

Nothing fitted: beats no-velocity baseline, 6/6 wearers

What it achieved, and what's next

~2/3

walking motion removed on hardware

Next: faster estimator

- Frames, calibration → camera extrinsics
- Latency budget → sensor time-sync
- Ground-truth the metric

Thank you. Questions?

Results, simulation, figures

github.com/fechachris4/msc_project

C++ real-time controller

github.com/fechachris4/HumanSL_MAIN

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